

SUGARSENSE

Early Diabetes Risk Screening Using Supervised Machine Learning

Project Report

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Project Type: Machine Learning / Data Science Mini Project

Domain: Healthcare / Machine Learning

Technology: Python, Pandas, NumPy, Scikit-learn, XGBoost, Streamlit

Medical Disclaimer: This project is developed for educational purposes only. It is not a medical diagnostic system and should not replace professional medical advice.

1. Abstract

SugarSense is a supervised machine learning project developed to predict whether a patient is likely to have diabetes based on clinical measurements. The project uses the Pima Indians Diabetes Database, containing 768 patient records and eight clinical input features. The target variable, Outcome, represents whether the patient is diabetic or non-diabetic.

The project follows a machine learning workflow including data auditing, identification of hidden missing values, exploratory data analysis, feature engineering, train-test splitting, preprocessing pipelines, model comparison, cross-validation, hyperparameter tuning, and model evaluation.

Six classification algorithms were considered: Logistic Regression, K-Nearest Neighbors, Decision Tree, Random Forest, XGBoost, and Support Vector Machine. Because this is a medical screening problem, Recall and ROC-AUC are especially important because incorrectly classifying a diabetic patient as non-diabetic can be more serious than generating a false alarm.

The cross-validation results showed that Logistic Regression achieved the highest mean ROC-AUC in the initial comparison, while Random Forest achieved the highest tuned cross-validation ROC-AUC and was selected as the final model according to the model-selection procedure.

A Streamlit-based interface is also included to provide an interactive prediction experience.

2. Introduction

Diabetes is a major chronic health condition that can remain undiagnosed for a long time. Early identification of people who may be at higher risk can encourage them to seek appropriate medical testing and professional healthcare advice.

Machine learning can be used to identify patterns in patient health measurements and estimate whether a patient may belong to a higher-risk group.

The objective of SugarSense is not to diagnose diabetes but to demonstrate how supervised machine learning can be used as an educational risk-screening tool.

3. Problem Statement

The project aims to build a supervised binary classification system that predicts whether a patient is likely to have diabetes.

Target	Meaning
Outcome = 0	Non-Diabetic
Outcome = 1	Diabetic

The model uses clinical measurements related to pregnancy history, glucose, blood pressure, skin thickness, insulin, BMI, diabetes family-history score, and age. The main challenge is to identify diabetic patients while reducing false-negative predictions.

4. Objectives

- Understand and audit the diabetes dataset.
- Identify hidden missing values represented by zero.
- Perform exploratory data analysis.
- Analyze relationships between clinical measurements and diabetes outcome.
- Create meaningful domain-based features.

- Build preprocessing pipelines without data leakage.
- Train multiple classification algorithms.
- Compare models using multiple evaluation metrics.
- Tune selected machine learning models.
- Select the final model using cross-validation performance.
- Develop an interactive Streamlit application.
- Communicate limitations and responsible use of the model.

5. Dataset Description

The project uses the Pima Indians Diabetes Database. The dataset contains 768 patient records, eight input features, and one target variable. The original project specification describes the dataset as containing female patients aged 21 years or older.

Feature	Description
Pregnancies	Number of times the patient has been pregnant
Glucose	Plasma glucose concentration
BloodPressure	Diastolic blood pressure
SkinThickness	Triceps skin-fold thickness
Insulin	2-hour serum insulin
BMI	Body Mass Index
DiabetesPedigreeFunction	Diabetes family-history related score
Age	Age of the patient
Outcome	Target: 0 = non-diabetic, 1 = diabetic

6. Dataset Audit

The first stage of the project was to inspect the structure and quality of the dataset. The audit included dataset shape, data types, descriptive statistics, duplicate records, class distribution, detection of impossible values, and percentage of problematic values.

The dataset contains 768 rows and no duplicate records.

Outcome	Number of Patients	Percentage
0 – Non-Diabetic	500	65.10%
1 – Diabetic	268	34.90%

7. Identification of Hidden Missing Values

Some missing clinical measurements are represented by 0 rather than actual NaN values. For example, a glucose value of zero is not a realistic clinical measurement. These values were treated as missing rather than genuine measurements.

Feature	Problematic Values	Percentage
Glucose	5	0.65%

Feature	Problematic Values	Percentage
BloodPressure	35	4.56%
SkinThickness	227	29.56%
Insulin	374	48.70%
BMI	11	1.43%

The project uses median imputation inside preprocessing pipelines so that imputation statistics are learned from the appropriate training data rather than from the complete dataset.

8. Exploratory Data Analysis

Exploratory Data Analysis was performed to understand feature distributions and relationships with diabetes outcome. The analysis included feature distribution plots, diabetic versus non-diabetic comparisons, class distribution visualization, correlation analysis, correlation heatmap, and problematic-value visualization.

The strongest simple correlation with diabetes outcome was observed for Glucose. Other useful variables included BMI, Age, Pregnancies, and DiabetesPedigreeFunction. BloodPressure and SkinThickness showed comparatively weaker linear relationships.

Feature	Correlation with Outcome
Glucose	0.467
BMI	0.293
Age	0.238
Pregnancies	0.222
DiabetesPedigreeFunction	0.174

9. Feature Engineering

Feature engineering was performed to provide the models with additional information derived from existing patient measurements.

9.1 Obesity Flag

A binary feature was created from BMI to indicate whether BMI is at least 30.

9.2 Glucose-BMI Interaction

$\text{Glucose_BMI} = \text{Glucose} \times \text{BMI}$. This combines glucose level and BMI into an interaction feature.

9.3 Age-Glucose Interaction

$\text{Age_Glucose} = \text{Age} \times \text{Glucose}$. This combines age and glucose information.

The finalized notebook also includes additional row-level feature engineering and missing-measurement indicators.

10. Data Splitting

The dataset was divided into training and testing sets using an 80/20 split. A stratified split was used so that the proportion of diabetic and non-diabetic cases remained reasonably consistent between the training and testing datasets. `random_state = 42` was used for reproducibility.

The test set is intended to remain untouched during model selection and tuning.

11. Data Preprocessing

The finalized workflow uses preprocessing pipelines to reduce data leakage. For models that require scaling, the workflow follows imputation, standard scaling, and model fitting. Scaling is particularly relevant for Logistic Regression, KNN, and SVM. Tree-based models do not require feature scaling in the same way.

Imputation is placed inside the model pipeline so that the imputer learns only from the relevant training data or fold.

12. Machine Learning Models

- Logistic Regression
- K-Nearest Neighbors (KNN)
- Decision Tree
- Random Forest
- XGBoost
- Support Vector Machine (SVM)

13. Model Evaluation Metrics

Accuracy: Measures the overall percentage of correct predictions.

Precision: Measures how many patients predicted as diabetic were actually diabetic.

Recall: Measures how many actual diabetic patients were correctly identified.

F1 Score: Combines precision and recall into a single score.

ROC-AUC: Measures how well the model separates diabetic and non-diabetic cases across different thresholds.

Overfit Gap: Compares training performance with validation performance to identify possible overfitting.

14. Five-Fold Cross-Validation Results

The finalized notebook used 5-fold cross-validation for model comparison.

Model	Accuracy	Precision	Recall	F1	ROC-AUC	Train AUC	Gap
Logistic Regression	0.7687	0.6468	0.7430	0.6912	0.8401	0.8604	0.0203
SVM	0.7443	0.6160	0.7196	0.6627	0.8371	0.9127	0.0757
Random Forest	0.7573	0.6788	0.5888	0.6278	0.8290	1.0000	0.1710
KNN	0.7573	0.7067	0.5185	0.5962	0.8225	1.0000	0.1775
XGBoost	0.7507	0.6314	0.6960	0.6605	0.8179	0.9964	0.1786
Decision Tree	0.7035	0.5600	0.7431	0.6344	0.7317	0.9251	0.1934

Logistic Regression achieved the highest initial mean ROC-AUC of 0.8401. Several models showed larger train-validation gaps, suggesting greater risk of overfitting.

15. Hyperparameter Tuning

Three models were tuned using RandomizedSearchCV with ROC-AUC as the scoring metric.

Model	Tuned CV ROC-AUC
Logistic Regression	0.8428
Random Forest	0.8448
SVM	0.8425

Random Forest achieved the highest tuned cross-validation ROC-AUC of 0.8448. Therefore, Random Forest was selected as the final model according to the cross-validation-based model-selection procedure.

16. Threshold Tuning

A classification threshold of 0.5 is not necessarily ideal for a medical screening problem. The project targets a recall of at least 85% while obtaining the best possible precision under that requirement.

This approach is intended to reduce the number of diabetic patients incorrectly classified as non-diabetic. The default and tuned thresholds can then be compared to explain the trade-off between recall and precision.

17. Final Model

Based on the tuned cross-validation results, Random Forest was selected as the final model. The selection was based on cross-validation performance rather than choosing a model only because it performed best on a test set.

18. Streamlit Application

A Streamlit application was developed to provide an interactive interface for the project. The application includes patient information input, diabetes prediction, probability breakdown, risk-level display, risk-factor analysis, basic recommendations, a diabetes risk gauge, and a medical disclaimer.

The current uploaded app.py is a simplified implementation. It currently uses Logistic Regression and asks for four inputs: Blood Pressure, Skin Thickness, BMI, and Age. It displays non-diabetic and diabetic probabilities, risk level, risk factors, recommendations, and a gauge chart.

This means the current Streamlit application should be synchronized with the finalized notebook before being presented as the final deployed version.

19. Model Saving

The project uses Joblib to save model-related files for reuse by the Streamlit application. The current application expects:

- Diabetes_Disease_prediction.pkl
- scaler.pkl
- columns.pkl

The finalized notebook also contains a workflow for saving the trained classification pipeline and cleaned feature-engineered dataset.

20. Technology Stack

Category	Technologies
Programming	Python
Data Processing	Pandas, NumPy
Visualization	Matplotlib, Seaborn

Category	Technologies
Machine Learning	Scikit-learn, XGBoost
Web Application	Streamlit
Model Serialization	Joblib
Interactive Visualization	Plotly

21. Project Workflow

Raw Dataset → Data Audit → Problematic Value Detection → Data Cleaning → Exploratory Data Analysis → Feature Engineering → Stratified Train/Test Split → Preprocessing Pipelines → 5-Fold Cross-Validation → Model Comparison → Hyperparameter Tuning → Final Model Selection → Threshold Analysis → Model Saving → Streamlit Application → Diabetes Risk Screening

22. Advantages

- Uses multiple machine learning algorithms.
- Performs exploratory data analysis before modeling.
- Identifies hidden missing values.
- Uses preprocessing pipelines.
- Uses stratified cross-validation.
- Considers ROC-AUC and Recall instead of accuracy alone.
- Performs hyperparameter tuning.
- Considers the cost of false-negative predictions.
- Provides an interactive Streamlit interface.
- Includes a medical disclaimer.

23. Limitations

- **Dataset limitation:** The dataset represents a specific population and may not generalize equally to every population.
- **Dataset size:** The dataset contains 768 records, which is relatively small for a healthcare prediction system.
- **Missing measurements:** Several clinical measurements contain zero values representing missing or unavailable measurements.
- **Generalization:** Good cross-validation performance does not guarantee equal performance on completely new populations.
- **Application alignment:** The current Streamlit application uses Logistic Regression and four inputs, while the finalized notebook selects Random Forest and uses the full project workflow.

24. Ethical Considerations

The SugarSense model should be considered a screening aid rather than a diagnostic system. It should not replace a doctor, laboratory testing, clinical examination, or professional medical advice.

The model can produce false positives and false negatives. A false-negative prediction could incorrectly suggest that a diabetic patient is not at risk. Real-world use would require additional clinical validation, a more representative

dataset, medical oversight, and careful evaluation of fairness and safety.

25. Future Improvements

- Synchronize the Streamlit application with the finalized Random Forest model.
- Add all eight original clinical inputs to the Streamlit interface.
- Use the same feature-engineering process used during training.
- Add the final preprocessing pipeline directly to the deployed model.
- Display ROC and Precision-Recall curves.
- Add a confusion matrix visualization.
- Add feature-importance explanations.
- Improve threshold-based screening.
- Compare different imputation strategies.
- Add a learning curve.
- Perform additional external validation.
- Deploy the finalized version online.

26. Conclusion

SugarSense demonstrates how supervised machine learning can be applied to diabetes risk screening using clinical measurements.

The project began with data auditing and exploratory analysis, where hidden missing values represented by zero were identified. Feature engineering was then used to create additional information from existing patient measurements.

Multiple machine learning models were evaluated using cross-validation. Logistic Regression achieved the highest initial mean ROC-AUC of 0.8401, while hyperparameter tuning resulted in a best ROC-AUC of 0.8448 for Random Forest. Therefore, Random Forest was selected as the final model according to the cross-validation-based selection procedure.

The project also demonstrates the importance of considering Recall and ROC-AUC in healthcare screening rather than relying only on accuracy.

The Streamlit component provides an interactive interface for demonstrating diabetes-risk prediction. However, the current Streamlit implementation should be synchronized with the finalized notebook model before being considered the final deployed version.

Overall, SugarSense demonstrates an end-to-end educational machine learning workflow from data analysis and model development to interactive deployment.

27. References

- Pima Indians Diabetes Database.
- SugarSense – Early Diabetes Risk Screening with Supervised Machine Learning project specification.
- Scikit-learn documentation.
- XGBoost documentation.
- Streamlit documentation.

28. Final Medical Disclaimer

This project is created for educational purposes only. SugarSense is not a medical diagnostic tool. Predictions generated by the system should not be used as a substitute for professional medical advice, laboratory testing, or clinical diagnosis. Always consult a qualified healthcare professional for medical concerns.